JavaScript Data Manipulation Basics

■ 1. Variables

Variables are used to store data values. You can declare them with var, let, or const.

```
let name = "Adrian";   // String
let age = 23;          // Number
const pi = 3.1416;     // Constant value
```

■ 2. Objects

Objects hold data in key-value pairs. They are useful for representing structured data.

```
let person = {
  name: "Adrian",
  age: 23,
  isStudent: true
};

console.log(person.name);   // Access property → Adrian
person.age = 24;            // Modify property

let user = {
  username: "adrian123",
  greet: function() {
    return "Hello, " + this.username;
  }
};

console.log(user.greet()); // Hello, adrian123
```

■ 3. Arrays

Arrays store multiple values in an ordered list.

```
let fruits = ["apple", "banana", "mango"];

console.log(fruits[0]);    // apple
fruits.push("orange");     // Add element
fruits.pop();              // Remove last element
console.log(fruits.length); // Count items

fruits.forEach(function(fruit) {
  console.log(fruit);
});

for (let fruit of fruits) {
  console.log(fruit);
}
```

■ 4. Functions

Functions let you reuse blocks of code.

```
function greet(name) {
  return "Hello, " + name + "!";
}

console.log(greet("Adrian")); // Hello, Adrian!

const add = (a, b) => a + b;
console.log(add(5, 3)); // 8
```

■ Putting it all together

```
// Variables
let students = [];

// Function to add a student (Object) into array
function addStudent(name, age) {
  let student = { name: name, age: age };
  students.push(student);
}

// Adding data
addStudent("Adrian", 23);
addStudent("Benedict", 25);

// Display students
students.forEach(student => {
  console.log(student.name + " is " + student.age + " years old.");
});

/* Output:
Adrian is 23 years old.
Benedict is 25 years old.
*/
```